Chapter 14D: Integrated Reproductive Review, Diagnostic Algorithms, and Graph-RAG Relationship Tables

This final reproductive segment consolidates female and male symptom approaches, endocrine pattern recognition, infertility frameworks, STI syndromes, cancer screening, emergency red flags, and relationship-first statements for graph-based retrieval. It is designed to convert diagrams and algorithms into dense text that can be chunked into a knowledge graph without losing clinical links.

Chapter orientation

Chapters 14A, 14B, and 14C built the physiology and disease-specific details. Chapter 14D is intentionally integrative. A reproductive complaint is often not solved by memorizing one organ system. Amenorrhea can be pregnancy, hypothalamic stress, pituitary prolactin excess, ovarian failure, thyroid disease, PCOS, uterine outflow obstruction, medication effect, or chronic systemic illness. Testicular pain can be torsion, epididymitis, orchitis, trauma, hernia, stone-referred pain, or tumor hemorrhage. Infertility can be ovulatory, tubal, uterine, cervical, male endocrine, male testicular, obstructive, genetic, medication-related, or couple-level. The purpose of this chapter is to connect symptom patterns to first tests, dangerous diagnoses, and mechanism-level relationships.

Use this chapter as a review map. The tables use compact wording, but the surrounding paragraphs spell out the causal logic: hormone pattern causes phenotype, infection causes inflammation, structural disease distorts anatomy, malignancy causes red flags, and screening tests are used only in defined risk groups rather than as universal diagnostic tests.

Graph-RAG principle: every row should be interpreted as an edge candidate. For example, chronic anovulation -> unopposed estrogen -> endometrial proliferation -> endometrial hyperplasia -> endometrial cancer risk. PID -> tubal inflammation -> scarring -> infertility and ectopic pregnancy risk. Diabetes and atherosclerosis -> endothelial dysfunction -> erectile dysfunction. Varicocele -> increased scrotal temperature and oxidative stress -> impaired spermatogenesis -> abnormal semen analysis.

1. Universal reproductive diagnostic rules

Several reproductive rules prevent missed diagnoses. First, pregnancy status changes the differential diagnosis, imaging choices, medication safety, and urgency of pelvic pain or bleeding. In any reproductive-age patient with amenorrhea, abnormal uterine bleeding, pelvic pain, syncope, or suspected ectopic pregnancy, a pregnancy test is the first diagnostic branch. Second, acute ischemic or obstructive emergencies need time-sensitive recognition: ovarian torsion and testicular torsion are clinical emergencies even when imaging is pending. Third, postmenopausal bleeding is endometrial cancer until proven otherwise. Fourth, painless testicular mass is testicular cancer until proven otherwise. Fifth, urethritis, cervicitis, PID, epididymitis, and genital ulcer disease require partner-aware STI thinking because treating only one patient can leave the source of reinfection untreated.

Clinical situation	First branch or first test	Dangerous diagnosis not to miss	Mechanistic reason
Reproductive-age pelvic pain	Urine or serum hCG before anchoring on gynecologic causes	Ectopic pregnancy, ovarian torsion, PID with tubo-ovarian abscess	Pregnancy and adnexal torsion change urgency and treatment; PID can scar tubes and cause sepsis.
Secondary amenorrhea	Pregnancy test, then TSH/prolactin/FSH/estradiol guided by history	Pregnancy, prolactinoma, ovarian insufficiency, hypothalamic suppression	Amenorrhea reflects interruption of the hypothalamic-pituitary-ovarian-uterine outflow axis.
Postmenopausal bleeding	Endometrial evaluation using transvaginal ultrasound and/or biopsy	Endometrial carcinoma	Bleeding after ovarian estrogen withdrawal is abnormal and may reflect malignant endometrial proliferation.

Clinical situation	First branch or first test	Dangerous diagnosis not to miss	Mechanistic reason
Acute testicular pain	Immediate torsion assessment; ultrasound should not delay surgery when suspicion is high	Testicular torsion	Twisting of the spermatic cord obstructs venous and arterial flow and threatens testicular viability.
Painless testicular mass	Scrotal ultrasound and tumor markers after diagnosis is suspected	Germ cell tumor	Intratesticular solid masses are malignant until proven otherwise.
Urethral discharge or cervicitis	NAAT for chlamydia/gonorrhea and empiric treatment when indicated	Ascending infection, PID, epididymitis, infertility	Mucosal infection can ascend through reproductive ducts and generate scarring.

2. Female symptom algorithms in words

Amenorrhea algorithm

Secondary amenorrhea means the patient previously menstruated and then stopped. The first rule is to test for pregnancy. If pregnancy is absent, the algorithm becomes an axis-localization problem. A low or normal FSH with low estradiol suggests central hypogonadism, often from functional hypothalamic amenorrhea, chronic illness, excessive exercise, undernutrition, opioid exposure, or pituitary disease. A high FSH with low estradiol suggests primary ovarian insufficiency or menopause because the pituitary is appropriately trying to stimulate a failing ovary. High prolactin suppresses GnRH and therefore lowers LH and FSH, producing anovulation, infertility, galactorrhea, and low estrogen effects. Abnormal TSH can disrupt ovulation through hypothalamic-pituitary effects and by changing sex hormone binding globulin and prolactin physiology. Hyperandrogenic amenorrhea points toward PCOS, androgen-secreting tumors, nonclassic congenital adrenal hyperplasia, Cushing syndrome, or medication exposure.

Amenorrhea pattern	Likely axis location	Common causes	Clinical associations
hCG positive	Pregnancy	Intrauterine pregnancy, ectopic pregnancy, early pregnancy loss	Nausea, breast tenderness, pelvic pain or bleeding if ectopic/loss.
Low/normal FSH + low estradiol	Hypothalamic or pituitary	Functional hypothalamic amenorrhea, pituitary mass, chronic disease, undernutrition, excessive exercise, opioids	Low libido, low bone density, stress, weight loss, headaches or visual symptoms if mass.
High FSH + low estradiol	Ovarian	Primary ovarian insufficiency, menopause, gonadal dysgenesis, chemo/radiation, autoimmune ovarian failure	Vasomotor symptoms, infertility, vaginal dryness, low bone density.
High prolactin	Pituitary or medication effect	Prolactinoma, antipsychotics, metoclopramide, hypothyroidism, chest wall stimulation	Galactorrhea, low libido, headaches, visual field defects if macroadenoma.
Hyperandrogenism	Ovary or adrenal androgen excess	PCOS, ovarian/adrenal tumor, nonclassic CAH, Cushing syndrome	Hirsutism, acne, oligomenorrhea, virilization if severe/rapid.
Normal hormones with cyclic pain but no bleeding	Outflow tract	Asherman syndrome, cervical stenosis, obstructive anomalies	History of uterine instrumentation, infection, or congenital anomaly.

Abnormal uterine bleeding algorithm

Abnormal uterine bleeding should be sorted by pregnancy status, hemodynamic stability, age, medication exposure, and structural versus nonstructural causes. PALM-COEIN is a useful classification: polyps, adenomyosis, leiomyomas, malignancy/hyperplasia, coagulopathy, ovulatory dysfunction, endometrial causes, iatrogenic causes, and not otherwise classified. Heavy bleeding since menarche suggests a bleeding disorder. Irregular unpredictable bleeding with skipped cycles suggests anovulation. Heavy regular bleeding with bulk symptoms suggests fibroids or adenomyosis. Intermenstrual or postcoital bleeding suggests cervical pathology, cervicitis, polyps, or malignancy. Postmenopausal bleeding requires endometrial evaluation because cancer risk rises when bleeding occurs after the endometrium should be inactive.

Bleeding clue	Pattern interpretation	Important causes	Next-step logic
Heavy bleeding since menarche	Coagulopathy signal	von Willebrand disease, platelet disorders, anticoagulants	Ask about epistaxis, bruising, dental/surgical bleeding; consider CBC and coagulation testing.
Irregular skipped cycles	Anovulatory bleeding	PCOS, thyroid disease, prolactin excess, perimenopause, hypothalamic dysfunction	Anovulation creates unpredictable estrogen exposure and unstable endometrium.
Heavy regular menses with pelvic pressure	Structural uterine disease	Leiomyomas, adenomyosis	Use pelvic exam and imaging; submucosal fibroids most strongly cause bleeding.
Intermenstrual/postcoital bleeding	Cervical or focal lesion signal	Cervicitis, cervical dysplasia/cancer, polyps	Evaluate cervix, STI risk, and screening status.
Bleeding after menopause	Cancer until proven otherwise	Endometrial hyperplasia/carcinoma, atrophy, polyps	Proceed to transvaginal ultrasound and/or endometrial sampling based on risk and findings.

Pelvic pain algorithm

Pelvic pain can be acute, chronic, cyclic, or pregnancy-associated. Acute severe unilateral pain with nausea and adnexal tenderness suggests ovarian torsion, especially with an ovarian cyst or mass. Pelvic pain plus fever, cervical motion tenderness, uterine tenderness, adnexal tenderness, or mucopurulent discharge suggests PID. Cyclic pain with dysmenorrhea, dyspareunia, dyschezia, or infertility suggests endometriosis. Diffuse pain with urinary symptoms may reflect cystitis or nephrolithiasis; diffuse abdominal pain can be appendicitis, inflammatory bowel disease, diverticulitis, or bowel obstruction. Chronic pelvic pain is often multifactorial and may combine endometriosis, pelvic floor dysfunction, bladder pain syndrome, IBS, trauma, and mood disorders.

Pain pattern	Disease pattern	High-yield associated findings	Why it matters
Sudden severe unilateral pelvic pain	Ovarian torsion, ruptured cyst, ectopic pregnancy	Nausea/vomiting, adnexal mass, positive hCG if ectopic	Torsion and ectopic pregnancy are time-sensitive emergencies.
Pelvic pain + fever/discharge	PID or tubo-ovarian abscess	Cervical motion tenderness, adnexal tenderness, mucopurulent cervicitis	Ascending infection can cause infertility, ectopic pregnancy, chronic pain, and sepsis.
Cyclic pain worsening before menses	Endometriosis	Dysmenorrhea, deep dyspareunia, dyschezia, infertility	Ectopic endometrial-like tissue drives inflammatory adhesions and pain.
Heavy painful enlarged uterus	Adenomyosis	Multiparity, boggy enlarged uterus, menorrhagia	Endometrial glands in myometrium cause uterine enlargement and painful bleeding.
Pelvic pain + urinary urgency/frequency	Cystitis, bladder pain syndrome, stone	Dysuria, hematuria, suprapubic pain	Urinary and reproductive symptoms overlap anatomically and neurologically.

3. Male symptom algorithms in words

Acute scrotum algorithm

Acute scrotal pain must be triaged for testicular torsion before less urgent causes. Torsion classically presents with sudden severe unilateral testicular pain, high-riding testis, abnormal lie, nausea/vomiting, and absent cremasteric reflex, although no single sign is perfect. Doppler ultrasound can support diagnosis, but surgical exploration should not be delayed when clinical suspicion is high. Epididymitis more often has gradual onset, posterior epididymal tenderness, urinary symptoms, fever, or STI exposure; Prehn sign is not reliable enough to rule out torsion. Torsion of the appendix testis can cause focal upper-pole tenderness and sometimes a blue-dot sign. Hernia, trauma, testicular tumor hemorrhage, and referred ureteral stone pain also belong in the acute scrotum differential.

Condition	Typical onset	Exam clues	Testing/treatment logic
Testicular torsion	Sudden, severe, often with nausea/vomiting	High-riding testis, transverse lie, absent cremasteric reflex	Urgent urologic evaluation; do not delay surgery for imaging if suspicion is high.

Condition	Typical onset	Exam clues	Testing/treatment logic
Epididymitis	Gradual posterior scrotal pain	Epididymal tenderness, urinary symptoms, fever, urethritis risk	NAAT for STI risk, urine studies; antibiotics based on age/risk and enteric vs STI pathogens.
Torsion appendix testis	Acute focal pain	Upper-pole tenderness, blue-dot sign sometimes	Supportive care after torsion excluded.
Incarcerated hernia	Acute groin/scrotal pain	Groin mass, bowel symptoms, nonreducible mass	Surgical evaluation if incarcerated/strangulated.
Testicular tumor with hemorrhage	Pain superimposed on mass	Palpable intratesticular mass	Scrotal ultrasound and tumor-marker evaluation.

Lower urinary tract symptoms and prostate algorithm

Older men with hesitancy, weak stream, nocturia, incomplete emptying, frequency, and urgency often have bladder outlet obstruction from benign prostatic hyperplasia. The prostate surrounds the urethra, so enlargement increases resistance to urinary flow; the bladder initially compensates with detrusor hypertrophy and later may decompensate with retention, hydronephrosis, recurrent infections, bladder stones, or renal dysfunction. Acute bacterial prostatitis presents with fever, pelvic or perineal pain, urinary symptoms, and a tender boggy prostate; vigorous prostate massage should be avoided because bacteremia can occur. Chronic prostatitis/chronic pelvic pain syndrome is pain-predominant and often culture-negative. Prostate cancer may be asymptomatic, detected through PSA screening, or present later with bone pain, weight loss, obstructive symptoms, or abnormal digital rectal exam.

Presentation	Most likely category	Key clues	Relationship statement
Gradual weak stream/nocturia	BPH	Age-related prostate enlargement, enlarged smooth prostate	Prostate enlargement -> urethral compression -> bladder outlet obstruction -> LUTS.
Fever + dysuria + perineal pain	Acute bacterial prostatitis	Tender boggy prostate, systemic illness	Ascending urinary infection -> prostate inflammation -> pain, fever, urinary obstruction risk.
Chronic pelvic/perineal pain	Chronic prostatitis/chronic pelvic pain syndrome	Pain >3 months, variable urinary/sexual symptoms	Neuromuscular and inflammatory mechanisms may persist without active bacterial infection.
Elevated PSA or abnormal DRE	Prostate cancer evaluation	Often asymptomatic; risk increases with age/family history	Malignant glandular proliferation -> PSA elevation and possible local/metastatic disease.
Acute urinary retention	Obstructive emergency	Painful inability to void	Outlet obstruction or medication effect -> bladder overdistension -> postrenal AKI risk.

4. Endocrine laboratory pattern recognition

Reproductive endocrinology is most manageable when labs are interpreted as feedback loops. The hypothalamus secretes pulsatile GnRH, the pituitary releases LH and FSH, and gonads produce sex steroids and inhibin. If the gonad fails, sex steroid feedback falls and pituitary gonadotropins rise. If the hypothalamus or pituitary fails, LH and FSH are inappropriately low or normal despite low sex steroids. If prolactin is high, GnRH pulsatility falls. If androgen excess is ovarian, adrenal, or tumoral, the speed of onset and severity of virilization matter. If testosterone is low, the distinction between primary testicular failure and secondary central hypogonadism changes the workup.

Lab pattern	Interpretation	Female examples	Male examples
High FSH/LH + low estradiol/testosterone	Primary gonadal failure	Primary ovarian insufficiency, menopause, Turner syndrome, chemo/radiation injury	Klinefelter syndrome, orchitis, chemo/radiation, testicular trauma.
Low/normal FSH/LH + low sex steroids	Central hypogonadism	Functional hypothalamic amenorrhea, pituitary mass, chronic disease, opioid effect	Pituitary disease, obesity-related functional suppression, opioids, anabolic steroid withdrawal.
High prolactin + low gonadotropins	Prolactin-mediated GnRH suppression	Amenorrhea, galactorrhea, infertility	Low libido, erectile dysfunction, infertility, gynecomastia.

Lab pattern	Interpretation	Female examples	Male examples
High testosterone or DHEA-S in female patient	Androgen excess localization	PCOS if mild/moderate; adrenal source suggested by high DHEA-S; tumor if rapid/severe	Not usually used as a male disease signal unless exogenous androgen exposure suspected.
Low AMH or high day-3 FSH in infertility context	Reduced ovarian reserve	Age-related decline, primary ovarian insufficiency	Not a male test.
Low semen count + high FSH	Impaired spermatogenesis with pituitary compensation	N/A	Primary testicular spermatogenic failure.
Low semen count + low/normal FSH/LH	Central or obstructive/functional pattern	N/A	Hypogonadotropic hypogonadism or nonendocrine infertility depending on semen/testosterone context.

5. STI syndrome table and consequences

STIs should be represented in a graph as pathogen-syndrome-complication relationships. Chlamydia and gonorrhea can cause cervicitis, urethritis, PID, epididymitis, disseminated infection, neonatal conjunctivitis, and infertility through scarring. Trichomonas causes vaginitis and urethritis and increases risk of HIV acquisition. Syphilis changes stage over time and can involve the skin, mucosa, nervous system, cardiovascular system, pregnancy, and fetus. HSV causes painful recurrent vesicles and ulcers, with severe neonatal risk when transmitted during delivery. HPV causes genital warts through low-risk types and anogenital cancers through high-risk oncogenic types. Partner treatment and retesting are part of disease control, not administrative add-ons.

Syndrome	Core pathogens	Typical findings	Complications/relationships
Cervicitis	Chlamydia trachomatis, Neisseria gonorrhoeae, Mycoplasma genitalium	Mucopurulent discharge, friable cervix, postcoital bleeding, pelvic discomfort	Cervicitis -> ascending infection -> PID -> tubal scarring -> infertility/ectopic pregnancy.
Urethritis	Chlamydia, gonorrhea, M. genitalium, trichomonas	Dysuria, urethral discharge, urethral irritation	Urethritis can ascend to epididymitis or prostatitis and can reinfect partners.
Vaginitis	BV flora, Candida, Trichomonas	Discharge, odor, pruritus, irritation; pH/microscopy help distinguish	BV and trichomonas increase susceptibility to other STIs and pregnancy complications.
PID	Polymicrobial; often chlam ydia/gonorrhea-associated	Pelvic pain, cervical motion/uterine/adnexal tenderness, fever/discharge possible	PID -> fallopian tube inflammation -> adhesions -> infertility, ectopic pregnancy, chronic pelvic pain.
Genital ulcer disease	HSV, syphilis; less commonly chancroid, LGV, granuloma inguinale	Painful vesicles/ulcers suggest HSV; painless chancre suggests primary syphilis	Ulcers disrupt mucosa and increase HIV transmission risk; syphilis progresses systemically if untreated.
Epididymitis	STI pathogens in younger/sexual-risk patients; enteric organisms in older/instrumented patients	Posterior testicular pain, epididymal tenderness, dysuria or discharge	Ascending urethral infection -> epididymal inflammation -> pain and potential fertility effects.
HPV disease	Low-risk HPV 6/11; high-risk HPV 16/18 and others	Warts or asymptomatic infection; dysplasia on screening	Persistent high-risk HPV -> cervical, anal, penile, vulvar, vaginal, and oropharyngeal cancers.

6. Infertility framework for both partners

Infertility should not be framed as a female-only disease. A useful graph model has three layers: production of gametes, transport of gametes, and implantation/gestation. Female infertility can result from ovulatory dysfunction, diminished ovarian reserve, tubal disease, endometriosis, uterine cavity distortion, cervical factors, or systemic disease. Male infertility can result from abnormal sperm production, impaired sperm function, obstruction, ejaculatory dysfunction, endocrine disease, varicocele, genetic conditions, infection, heat, medications, toxins, or prior surgery. Both partners should be evaluated in parallel when clinically appropriate because sequential testing can delay diagnosis and treatment.

Infertility domain	Female pathway	Male pathway	Graph-RAG edge examples
Gamete production	Ovulatory dysfunction, PCOS, hypothalamic amenorrhea, primary ovarian insufficiency, aging	Primary testicular failure, varicocele, genetic disorders, chemo/radiation, anabolic steroids	PCOS -> anovulation -> infertility; varicocele -> impaired spermatogenesis -> oligospermia.
Hormonal signaling	GnRH/LH/FSH/estradiol/prolactin/thyroid disruption	GnRH/LH/FSH/testosterone/prolactin disruption	Hyperprolactinemia -> GnRH suppression -> low LH/FSH -> infertility.
Transport anatomy	Tubal scarring from PID/endometriosis/surgery	Vas deferens obstruction, ejaculatory duct obstruction, retrograde ejaculation	PID -> tubal scarring -> infertility and ectopic pregnancy.
Uterine/cavity environment	Fibroids, polyps, adhesions, congenital anomalies, endometritis	N/A	Submucosal fibroid -> cavity distortion -> implantation failure or bleeding.
Sexual function	Dyspareunia, vaginismus, endocrine low libido	Erectile dysfunction, ejaculatory dysfunction, libido disorders	Diabetes -> endothelial/neuropathy effects -> erectile dysfunction -> infertility risk.
Systemic/iatrogenic factors	Autoimmune disease, renal disease, medications, obesity, undernutrition	Medications, toxins, heat, obesity, systemic illness	Cyclophosphamide -> gonadal toxicity -> infertility risk.

Semen analysis interpretation

Semen analysis is the entry point for male-factor infertility, but it is not the final diagnosis. Abnormal volume can suggest collection issues, ejaculatory duct obstruction, retrograde ejaculation, androgen deficiency, or congenital absence of the vas deferens. Low sperm concentration can reflect impaired production, endocrine disease, varicocele, genetic causes, infection, medication toxicity, heat exposure, or obstruction. Poor motility can reflect varicocele, antisperm antibodies, infection, oxidative stress, or structural flagellar disorders. Abnormal morphology is nonspecific but can contribute to impaired fertilization. Azoospermia requires separation into obstructive versus nonobstructive patterns using history, exam, semen volume, FSH, testicular size, and sometimes genetic testing or imaging.

Semen finding	Meaning	Common causes	Next interpretation step
Low volume	Emission/collection/obstruction issue	Incomplete collection, retrograde ejaculation, ejaculatory duct obstruction, androgen deficiency	Repeat test; consider post-ejaculatory urine, pH/fructose, obstruction evaluation.
Oligospermia	Low sperm concentration	Varicocele, endocrine disease, testicular failure, toxins, heat, medications	Check endocrine profile and exam; consider reversible exposures.
Azoospermia + high FSH/small testes	Nonobstructive testicular failure	Genetic causes, prior chemo/radiation, orchitis, primary testicular failure	Consider karyotype/Y microdeletion testing depending on severity.
Azoospermia + normal FSH/normal testes	Obstructive or ejaculatory cause	Vas deferens absence, vasectomy, infection scarring, ejaculatory duct obstruction	Assess volume, pH, fructose, vas deferens, imaging if indicated.
Asthenozoospermia	Poor motility	Varicocele, infection, oxidative stress, sperm structural defects	Repeat semen analysis; evaluate infection/varicocele/exposures.

7. Cancer screening and diagnostic distinction

Screening tests are used in asymptomatic defined-risk populations; diagnostic tests evaluate symptoms, masses, bleeding, abnormal screening results, or red flags. A mammogram is screening in an asymptomatic age-eligible patient, but a palpable breast mass requires diagnostic imaging and tissue diagnosis when suspicious. Pap and HPV testing are screening for cervical precancer, but visible cervical lesions, postcoital bleeding, or abnormal discharge require diagnostic evaluation regardless of screening schedule. PSA screening is shared decision-making rather than a universal command; elevated PSA can reflect cancer, BPH, prostatitis, recent ejaculation, instrumentation, or urinary retention. Testicular cancer is usually detected by symptom or exam rather than population screening; a painless intratesticular mass should be evaluated with scrotal ultrasound.

Cancer/domain	Screening or detection logic	Red flags	Core relationships
Cervical cancer	Age-based cytology/HPV screening for average-risk patients	Postcoital bleeding, visible cervical lesion, abnormal bleeding	Persistent high-risk HPV -> CIN -> invasive cervical cancer.

Cancer/domain	Screening or detection logic	Red flags	Core relationships
Breast cancer	USPSTF: biennial mammography for women 40-74 at average risk	Palpable mass, nipple retraction, bloody discharge, skin dimpling, axillary nodes	Malignant breast epithelial proliferation -> mass/calcifications -> local invasion/nodal spread.
Endometrial cancer	No routine screening for average risk; evaluate abnormal bleeding	Postmenopausal bleeding, chronic anovulation, obesity, Lynch syndrome	Unopposed estrogen -> endometrial hyperplasia -> endometrioid carcinoma risk.
Ovarian cancer	No routine screening for average-risk asymptomatic patients	Bloating, early satiety, pelvic/abdominal mass, ascites, family history	Often late presentation because early ovarian cancer symptoms are nonspecific.
Prostate cancer	PSA screening by shared decision-making in appropriate age/risk groups	Bone pain, weight loss, obstructive symptoms, abnormal DRE	Prostate adenocarcinoma -> PSA elevation and possible osteoblastic bone metastases.
Testicular cancer	No population screening; evaluate masses promptly	Painless intratesticular mass, heaviness, gynecomastia, back pain	Germ cell tumor -> testicular mass -> tumor markers/metastatic spread depending on subtype.

8. Contraception, pregnancy-aware medicine, and contraindication review

Contraceptive choice is a risk-benefit decision that links reproductive goals, thrombotic risk, bleeding pattern, contraindications, medication interactions, and noncontraceptive benefits. Estrogen-containing methods suppress ovulation and stabilize bleeding but increase venous thromboembolism risk, so they are avoided in major estrogen contraindications such as current breast cancer, migraine with aura, severe uncontrolled hypertension, high-risk thrombophilia, prior VTE in high-risk settings, smoking age 35 or older with significant tobacco use, and early postpartum high-risk windows. Progestin-only methods are often safer when estrogen is contraindicated, but each method has distinct bleeding, bone, metabolic, or procedural considerations. IUDs provide highly effective long-acting contraception; copper IUDs are hormone-free but can worsen bleeding and cramps, whereas levonorgestrel IUDs reduce bleeding and endometrial proliferation.

Method class	Primary mechanism	Major advantages	Major cautions
Combined hormonal contraception	Suppresses ovulation; thickens cervical mucus; stabilizes endometrium	Cycle control, acne/PCOS symptom benefit, reduced ovarian/endometrial cancer risk	Avoid with major estrogen contraindications; VTE and stroke risk depend on patient factors.
Progestin-only pill/implant/injection	Thickens mucus; variable ovulation suppression; endometrial atrophy	Estrogen-free option; implant highly effective	Irregular bleeding; DMPA can reduce bone density and delay return to fertility.
Levonorgestrel IUD	Local progestin effect thickens mucus and thins endometrium	Long-acting, reduces heavy menstrual bleeding, low systemic hormone	Insertion procedure; irregular bleeding early; rare perforation/expulsion.
Copper IUD	Copper creates sperm-toxic inflammatory uterine environment	Hormone-free, long-acting, can be emergency contraception	May worsen bleeding/cramping; avoid with active pelvic infection.
Emergency contraception	Delays ovulation or prevents fertilization depending on method	Time-sensitive pregnancy prevention after unprotected intercourse	Effectiveness varies by method, timing, body weight, and ovulation timing.

Medication and pregnancy-aware edges

Pregnancy-aware medicine should be encoded as medication-condition-fetal risk relationships. ACE inhibitors and angiotensin receptor blockers can injure the fetal renal system, especially after the first trimester. Isotretinoin is strongly teratogenic and requires strict pregnancy prevention. Valproate increases neural tube defect and neurodevelopmental risk. Warfarin can cause fetal embryopathy and bleeding risk, although special situations such as mechanical valves require expert management. Methotrexate is contraindicated in pregnancy because it antagonizes folate and can cause embryopathy and pregnancy loss. Tetracyclines can affect teeth and bone development later in pregnancy. The clinical action is not simply to memorize lists, but to connect reproductive potential, pregnancy testing, contraception, and medication review before prescribing.

9. Emergency and red-flag recognition table

Red flag	Likely emergency or serious disease	Action logic	RAG relationship phrasing
Positive pregnancy test + pelvic pain/bleeding/syncope	Ectopic pregnancy until proven otherwise	Urgent evaluation with quantitative hCG and ultrasound pathway	Ectopic implantation -> tubal rupture risk -> hemorrhagic shock.
Sudden unilateral pelvic pain + vomiting	Ovarian torsion	Urgent gynecologic evaluation; ultrasound helps but clinical urgency matters	Adnexal twist -> vascular compromise -> ovarian necrosis.
Postmenopausal bleeding	Endometrial cancer or hyperplasia until proven otherwise	Endometrial assessment	Bleeding after menopause -> abnormal endometrial activity -> malignancy evaluation.
Severe pelvic pain + fever + adnexal mass	Tubo-ovarian abscess	Hospital-level evaluation, antibiotics, drainage/surgery depending on severity	PID -> abscess formation -> sepsis/rupture risk.
Acute testicular pain with high-riding testis	Testicular torsion	Immediate urologic evaluation	Spermatic cord torsion -> ischemia -> testicular loss.
Painless solid intratesticular mass	Testicular cancer	Ultrasound and urology referral	Intratesticular solid mass -> malignancy risk.
Erection >4 hours	Priapism	Emergency evaluation, especially ischemic priapism	Trapped deoxygenated blood -> corporal ischemia -> erectile dysfunction.
Breast mass with skin dimpling/nipple retraction	Breast cancer	Diagnostic imaging and biopsy when indicated	Malignancy -> stromal tethering -> skin/nipple changes.
Fever + tender boggy prostate	Acute bacterial prostatitis	Antibiotics; avoid vigorous massage	Prostate infection -> bacteremia risk with manipulation.

10. Differential diagnosis master tables

Amenorrhea differential by axis

Axis segment	Disease examples	Mechanism	Clinical clues
Hypothalamus	Functional hypothalamic amenorrhea, chronic illness, undernutrition, excessive exercise, stress	Reduced GnRH pulsatility -> low LH/FSH -> low estradiol -> anovulation	Weight loss, athletic training, stress, low BMI, low bone density.
Pituitary	Prolactinoma, Sheehan syndrome, pituitary mass, infiltrative disease	Impaired gonadotropin secretion or prolactin-mediated GnRH suppression	Galactorrhea, headaches, visual symptoms, failure to lactate after hemorrhage.
Ovary	PCOS, primary ovarian insufficiency, menopause, resistant ovary syndrome	Anovulation or follicle depletion -> estrogen/progesterone imbalance	Hyperandrogenism in PCOS; vasomotor symptoms/high FSH in ovarian insufficiency.
Uterus/outflow	Asherman syndrome, cervical stenosis, müllerian anomaly, imperforate hymen	Endometrium cannot shed or blood cannot exit despite hormonal cycling	Cyclic pain, prior curettage/infection, normal secondary sex characteristics.
Systemic/endocrine	Thyroid disease, Cushing syndrome, uncontrolled diabetes, CKD, medications	Systemic illness or medication disrupts HPO axis and metabolism	Other endocrine/systemic symptoms.

Hyperandrogenism differential

Cause	Onset/severity	Associated pattern	Distinguishing logic
PCOS	Gradual, mild to moderate	Oligo/anovulation, hirsutism, acne, metabolic syndrome	Most common; diagnosis requires excluding mimics and recognizing ovulatory dysfunction plus hyperandrogenism.
Androgen-secreting tumor	Rapid and severe	Virilization, very high testosterone or DHEA-S	Rapid onset or virilization is not typical PCOS.
Nonclassic CAH	Variable; often adolescence/adulthood	Hyperandrogenism, menstrual irregularity, infertility	Screen with 17-hydroxyprogesterone when suspected.
Cushing syndrome	Progressive systemic features	Central obesity, proximal weakness, striae, hypertension, diabetes	Cortisol excess can coexist with androgen excess.
Medication/anabolic exposure	Depends on exposure	Virilization or cycle disruption	Medication and supplement history is diagnostic.

Scrotal mass differential

Mass type	Pain?	Transillumination	Key clues	Clinical logic
Hydrocele	Usually painless	Yes	Fluid around testis, scrotal swelling	Fluid collection transmits light and is usually benign unless new/secondary.
Spermatocele	Usually painless	May transilluminate	Cystic epididymal head mass	Epididymal cystic lesion separate from testis.
Varicocele	Dull ache/heaviness	No	Bag of worms, worse standing/Valsalva, usually left-sided	Dilated pampiniform plexus -> heat/oxidative stress -> infertility risk.
Testicular tumor	Often painless	No	Solid intratesticular mass	Solid intratesticular mass is malignant until proven otherwise.
Epididymitis	Painful	No	Posterior tenderness, dysuria/discharge possible	Infection/inflammation of epididymis causes pain and swelling.
Inguinal hernia	Variable	No	Groin impulse, bowel sounds, reducibility	Abdominal contents descend through inguinal canal into scrotum.

11. Graph-RAG relationship tables

The following relationship tables are intentionally written as triples and causal chains. They are not meant to replace the narrative chapter; they are meant to give an extraction pipeline repeated edges with clear directionality. The graph should distinguish causes, complications, diagnostic indicators, contraindications, treatments, risk factors, and mimics.

Source concept	Relationship	Target concept	Justification phrase
Pregnancy	must be excluded before diagnosing	secondary amenorrhea	Pregnancy is the most common and highest-impact initial branch in reproductive-age amenorrhea.
Hyperprolactinemia	suppresses	GnRH pulsatility	Prolactin reduces hypothalamic GnRH signaling.
Reduced GnRH pulsatility	causes	low LH/FSH	The pituitary receives less pulsatile stimulation.
Low LH/FSH	causes	anovulation/hypogonadism	Gonads receive inadequate trophic stimulation.
PCOS	causes	chronic anovulation	Follicular dysfunction and hyperandrogenism disrupt regular ovulation.
Chronic anovulation	causes	unopposed estrogen exposure	Progesterone withdrawal is absent or infrequent.
Unopposed estrogen	increases risk of	endometrial hyperplasia/cancer	Persistent proliferative signaling can produce hyperplasia.
PID	causes	fallopian tube inflammation	Ascending infection reaches upper reproductive tract.
Fallopian tube inflammation	causes	tubal scarring	Inflammation heals with adhesions and fibrosis.

Source concept	Relationship	Target concept	Justification phrase
Tubal scarring	increases risk of	infertility and ectopic pregnancy	Gamete/embryo transport is impaired.

Source concept	Relationship	Target concept	Justification phrase
Varicocele	impairs	spermatogenesis	Venous dilation increases scrotal temperature and oxidative stress.
Testicular torsion	causes	testicular ischemia	Cord twisting compromises venous and arterial flow.
Testicular ischemia	can cause	testicular loss/infertility	Prolonged ischemia causes irreversible tissue injury.
Diabetes mellitus	contributes to	erectile dysfunction	Endothelial dysfunction, neuropathy, and vascular disease impair erection physiology.
Atherosclerosis	contributes to	erectile dysfunction	Penile vascular disease can precede overt coronary disease.
BPH	causes	bladder outlet obstruction	Enlarged transition-zone tissue compresses the urethra.
Bladder outlet obstruction	causes	urinary retention/hydronephrosis	High downstream resistance impairs bladder emptying and can transmit pressure upstream.
High-risk HPV	causes	cervical dysplasia/cancer	Persistent oncogenic HPV alters cell-cycle regulation.
BRCA1/BRCA2 pathogenic variants	increase risk of	breast and ovarian cancer	Defective homologous recombination increases malignant transformation risk.
Lynch syndrome	increases risk of	endometrial and ovarian cancer	Mismatch repair deficiency promotes malignancy in associated tissues.

12. Integrated cases for pattern recognition

Case stem	Most likely diagnosis/category	First diagnostic logic	Key mechanism
A 22-year-old runner with weight loss, low BMI, secondary amenorrhea, low estradiol, and low-normal FSH.	Functional hypothalamic amenorrhea	Exclude pregnancy; interpret low gonadotropin/low estrogen as central suppression.	Energy deficit/stress -> low GnRH pulsatility -> low LH/FSH -> anovulation.
A 31-year-old with oligomenorrhea, hirsutism, acne, obesity, and infertility.	PCOS	Exclude pregnancy and mimics; assess hyperandrogenism and ovulatory dysfunction.	Hyperandrogenism and follicular dysfunction -> chronic anovulation -> infertility/endometrial risk.
A 56-year-old with new vaginal bleeding after 5 years without menses.	Endometrial cancer/hyperplasia evaluation	Postmenopausal bleeding requires endometrial assessment.	Bleeding after menopause signals abnormal endometrial activity.
A 19-year-old with sudden severe unilateral testicular pain, vomiting, high-riding testis.	Testicular torsion	Urgent urologic evaluation; do not delay if high suspicion.	Cord twist -> ischemia -> testicular loss.
A 25-year-old with mucopurulent cervicitis, pelvic pain, cervical motion tenderness, and fever.	PID	Treat clinically and test for STIs; evaluate severity/abscess.	Ascending infection -> upper tract inflammation -> tubal scarring.
A 34-year-old man with infertility, left bag-of-worms scrotal mass worse standing, abnormal semen analysis.	Varicocele-associated male infertility	Confirm exam/ultrasound when indicated and evaluate semen/endocrine context.	Pampiniform dilation -> heat/oxidative stress -> impaired spermatogenesis.
A 68-year-old with nocturia, weak stream, hesitancy, enlarged smooth prostate.	BPH	Assess symptoms, urinalysis, retention/renal complications as needed.	Prostate enlargement -> urethral compression -> bladder outlet obstruction.
A 45-year-old with erectile dysfunction, diabetes, hypertension, and no morning erections.	Organic erectile dysfunction	Assess vascular/metabolic/endocrine/medication factors.	Endothelial dysfunction and neuropathy impair erection physiology.

13. Review checklist

• **Pregnancy first:** reproductive-age amenorrhea, pelvic pain, and abnormal bleeding require pregnancy status before narrowing the differential.

• **Axis localization:** high gonadotropins mean primary gonadal failure; low or inappropriately normal gonadotropins mean central suppression or pituitary/hypothalamic disease.

• **Prolactin matters:** prolactin excess suppresses GnRH and can cause amenorrhea, infertility, low libido, erectile dysfunction, and galactorrhea.

• **Anovulation is not benign:** chronic anovulation can cause infertility and unopposed estrogen exposure, increasing endometrial hyperplasia and cancer risk.

• **PID has delayed consequences:** PID damages fallopian tubes and increases infertility, ectopic pregnancy, and chronic pelvic pain risk.

• **Acute scrotum is torsion until proven otherwise:** testicular salvage depends on rapid recognition and urologic management.

• **Postmenopausal bleeding is cancer until proven otherwise:** evaluate the endometrium rather than reassuring based on intermittent symptoms.

• **Painless testicular mass is cancer until proven otherwise:** ultrasound and urology referral are core steps.

• **Screening is not diagnosis:** abnormal bleeding, masses, or visible lesions require diagnostic evaluation even if screening is up to date.

• **Infertility is couple-level:** evaluate ovulation, ovarian reserve, tubal/uterine anatomy, semen parameters, sexual function, endocrine factors, and systemic exposures.

References and guideline anchors

1. Centers for Disease Control and Prevention. Sexually Transmitted Infections Treatment Guidelines, 2021. CDC; 2021. Accessed May 2026.

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3. U.S. Preventive Services Task Force. Breast Cancer: Screening. Final Recommendation Statement. April 30, 2024. Accessed May 2026.

4. American Urological Association and American Society for Reproductive Medicine. Diagnosis and Treatment of Infertility in Men: AUA/ASRM Guideline. 2020; amended 2024. Accessed May 2026.

5. American Urological Association. Testosterone Deficiency Guideline. Accessed May 2026.

6. American College of Obstetricians and Gynecologists. Clinical guidance on contraception access, gynecologic symptoms, and reproductive health topics. Accessed May 2026.

7. International evidence-based guideline sources on PCOS and infertility were used as conceptual anchors for ovulatory dysfunction, hyperandrogenism, metabolic risk, and infertility workup frameworks.